VAST-GWO: a verified archive-guided sparse-evaluation grey wolf optimizer for global optimization and 3D UAV delivery path planning

Abstract

In black-box optimization the resource that decides solution quality is the number of real objective evaluations, not the number of arithmetic operations. The grey wolf optimizer (GWO) spends this resource uniformly over the whole pack and derives all movement from three leaders, which invites premature convergence and makes each evaluation less informative than it could be. This paper proposes VAST-GWO, a verified archive-guided sparse-evaluation GWO. In every generation, 100 cheap candidates are generated from three complementary routes: standard GWO leader moves (50), success-scale shells around the current alpha wolf (12), and multi-region shells and two-centre segments produced from a verified elite archive (38). Only $K = 10$ uniformly sampled candidates receive a real evaluation, and no unevaluated candidate ever influences leaders, archives or rankings—a verified-state discipline that keeps every piece of search memory backed by a billed evaluation. Across seven methods on CEC2017, VAST-GWO ranks first at both dimensions tested: at $D = 50$ it attains a mean rank of 2.24 with 100,000 true evaluations and 30 runs, significantly outperforming the 2026 hierarchical HAGWO, PSO, CLPSO, the 2026 adaptive oppositional AOGWO and standard GWO; at $D = 100$, with the same budget and 30 seeds, it attains a mean rank of 1.76 against six baselines, and against every one of them the Holm-corrected paired test returns more significant function-level wins than losses. A controlled equal-slot ablation shows that the verified archive carries the improvement: a GWO-only candidate pool loses 27 of the 29 functions to the full method, and replacing the archive with a single alpha-centred sampler with identical slots, scale and acceptance rule collapses the method to 2 wins. Directed screening, fixed quotas and three Q-learning schedulers do not improve on the uniform draw, indicating that the candidate-source structure, not the evaluation scheduler, is the decisive design variable. In a 209-dimensional, DEM-based 3D UAV delivery application with hard time windows, VAST-GWO is the only method that finds feasible routes in all 150 runs, attains the smallest censored mean time to feasibility, and produces the best paired final route quality against every baseline after Holm correction.

Keywords: grey wolf optimizer; sparse evaluation; verified elite archive; multi-region search; limited evaluation budget; UAV path planning; CEC2017.

1 Introduction

Nature-inspired optimizers are routinely used for engineering design, scheduling and path planning because they assume little about the objective and scale to black-box simulators [1–4]. The no-free-lunch theorems make the point precise: no single search strategy dominates across all objectives, so what matters is matching the structure of the search to the structure of the budget [5]. Their practical cost, however, is dominated by the objective function itself. In finite-element calibration, computational-fluid design or UAV mission planning a single evaluation may take seconds to hours, so the total number of *real* evaluations is the binding constraint rather than the number of arithmetic operations [6–11]. Under such budgets an optimizer has to answer one question well: *where should the next real evaluation be spent?*

The grey wolf optimizer (GWO) [12] is an attractive starting point. It is parameter-light, easy to implement and has been applied across the Internet of Things, feature selection, agriculture and UAV planning [13, 14]. Its classic form, however, evaluates the entire pack every generation and derives all movement from the three leaders α, β and δ. When evaluations are scarce this design has two structural consequences. First, the search state is thin: apart from the three leaders, the pack retains no memory of the regions it has visited, so a region that produced a verified improvement but no current leader is forgotten immediately. Second, the

movement distribution contracts around the same three attractors, which is the classical route to premature convergence and stagnation on multimodal landscapes. Both effects are visible in the behaviour of standard GWO on the CEC2017 suite and are documented across the GWO literature [13, 14].

A large recent literature improves GWO from different angles. Operator-level variants change the movement rule itself, adding differential evolution and elimination [15], mutation with evolutionary population dynamics [16], teaching–learning hybridisation [17], beetle-antennae search [18], grey wolf–particle swarm hybrids [19], self-repulsion [20] and engineering-oriented forms [21, 22]. Learning-driven variants let reinforcement or Q-learning choose strategies, leaders or budgets [23–26], and memory-based variants extend what the search retains [27, 28]. UAV-specific versions embed such mechanisms in trajectory planning [29, 30]. These methods deliver strong results, yet three gaps remain. Most of them are evaluated under protocols in which every individual is evaluated once per iteration, so the budget is expressed in iterations rather than in real evaluations and the two are not interchangeable. Most of them change the movement rule, so it is rarely isolated which design element produces the gain. And the natural alternative to a new mechanism—allocating the same budget differently, or simply enlarging the candidate set that the budget is drawn from—is seldom tested against matched controls.

This paper takes the real-evaluation budget itself as the design constraint and asks a deliberately narrow question: *given a fixed number of real evaluations per generation, how should the candidate pool be composed?* We propose VAST-GWO (verified archive-guided sparse-evaluation technique for grey wolf optimization), which separates cheap candidate generation from real evaluation and admits information into the search state only after it has been paid for by a real evaluation—a *verified-state discipline*. In every generation, $M = 100$ candidates are produced from three routes: 50 standard GWO leader moves, 12 success-scale shells around α, and 38 multi-region candidates generated from a verified elite archive. Exactly $K = 10$ candidates are drawn uniformly at random and evaluated; the evaluated subset is the only information that updates the leaders, the archive and the route statistics. Because the pool is never ranked, the composition of the pool is the design variable that matters, and it can be varied in isolation while the pool size, the evaluated subset size and the budget are held constant.

The hypotheses tested in this paper are three, and each is falsifiable with a matched control. **H1 (stagnation recovery).** A verified multi-region archive produces candidates in regions that the current leaders no longer represent, so VAST-GWO should leave stagnation episodes more frequently and spend less of its budget inside long stagnation than standard GWO. **H2 (exploration efficiency).** Because the pool contains verified information from several regions, a uniformly drawn evaluation should be more likely to open useful new ground, so VAST-GWO should achieve a lower final error at equal true evaluation count, with a larger advantage on simple-multimodal and hybrid functions than on unimodal ones. **H3 (scheduler redundancy).** If the candidate sources are the active ingredient, then elaborate allocation rules should not improve on a uniform draw once the pool is fixed—including a learned policy.

The evidence assembled in this paper supports H1, H2 and H3, and it does so with the same audited budget for every method. On CEC2017 at $D = 50$ with 100,000 true evaluations and 30 independent runs, VAST-GWO attains the best mean rank among seven methods (2.24), is significantly better than five of six baselines under Holm correction, and reaches statistical parity with the strong improved GWO while leading it on mean rank; at $D = 100$, with the same evaluation budget and 30 seeds, it attains the lowest mean rank against all six baselines, and it ranks first again under a budget five times smaller. A five-arm equal-slot ablation shows that the verified archive is the performance engine: a GWO-only pool loses 27 of the 29 functions to the full method, replacing the archive with a single alpha-centred sampler with identical slots, scale and acceptance rule collapses the method to 2 wins, and the alpha-shell route is optional once the archive is present. The boundary experiments reject directed screening, informed allocation, fixed quotas, stagnation-triggered radius boosting, success-direction steering and three Q-learning schedulers—the last only beating a random-quota control, never the best fixed policy. In a 209-dimensional DEM-based 3D UAV delivery problem with hard time windows and five real terrain tiles, VAST-GWO is the only method feasible in all 150 runs, attains the smallest censored mean time to feasibility, and produces the best paired final route quality against every baseline after Holm correction.

The remainder of this paper is organised as follows. Section 2 reviews related work and identifies the

research gap. Section 3 presents VAST-GWO. Section 4 describes the experimental protocol, Section 5 reports the optimization results and Section 6 the ablation and mechanism analysis. Section 7 presents the UAV application, Section 8 discusses implications and limitations, and Section 9 concludes.

2 Related work

2.1 Grey wolf optimizer and its variants

GWO [12] mimics the leadership hierarchy and cooperative hunting of grey wolves: the three best solutions guide the pack through encircling and attacking equations whose exploration–exploitation balance is controlled by a linearly decreasing parameter. Comprehensive reviews document its variants and applications in IoT, energy, scheduling and engineering design [13, 14]. The dominant line of research changes the movement equation: hybridisation with differential evolution and an elimination mechanism [15], adaptive differential-evolution schemes with external archives and success-history adaptation [31–33], mutation combined with evolutionary population dynamics and nonlinear population reduction [16], teaching–learning hybrids [17], beetle-antennae hybrids [18], particle swarm hybrids [19], self-repulsion strategies for feature selection [20] and improved forms for classical engineering design problems [21, 22]. A second line lets learning control the search: Q-learning or reinforcement learning selects strategies, leaders or budget shares [23–26, 34, 35], and UAV-specific variants embed such policies into trajectory planning [29, 30]. VAST-GWO belongs to neither line in its mechanism: it keeps the classical leader move for one route of the pool and changes *which candidates exist* under a fixed number of real evaluations per generation.

2.2 Memory and archive mechanisms in swarm intelligence

Memory is a recurring idea in population-based search, where it stabilises the search and preserves diversity. Elite and external archives are a mature device in evolutionary computation, used both to preserve non-dominated solutions [3] and to retain promising regions during search [31, 36, 37]. Recent GWO variants use elite memories, cross-search memories or two-archive structures for large-scale and multi-objective problems [27, 28]. VAST-GWO shares the motivation but differs in three respects. First, the archive is *verified*: only positions that received a real evaluation may enter, so the archive never stores a surrogate prediction or an unmeasured guess, and every archived value is a billed measurement. Second, the archive is *generative*: its centres and the segments between pairs of centres directly produce candidates, rather than serving only as a diversity reservoir or as a set of reference points. Third, it is *multi-region* by construction: a separation threshold prevents the archive from collapsing into a single cluster, so its candidates cover several verified basins at once. Section 6 shows that these properties are not decorative—the multi-region generative role, not the mere presence of extra candidates, produces the measured improvement.

2.3 Optimization under expensive evaluations

When evaluations are expensive, surrogate-assisted evolutionary algorithms are the dominant paradigm: a model of the objective guides the search and is updated with a small number of real evaluations [6–11]. VAST-GWO is complementary rather than competing. It uses no surrogate; it targets the regime in which a real evaluation is affordable but scarce enough that the per-generation allocation must be fixed and audited. This keeps the comparison clean—every method receives exactly the same number of billed evaluations and no method benefits from a model the others do not have—and it leaves the archive in a form that a future surrogate could use, because every entry is a real measurement.

2.4 UAV path planning with metaheuristics

Metaheuristics are widely used for three-dimensional UAV path planning, where objectives combine path length, energy or smoothness with hard constraints on terrain, obstacles, flight envelope and, in delivery

Table 1. Research lines in GWO and expensive optimization, and their relation to the present work.

Research line	Typical intervention	Relation to this work
GWO and its operator-level variants	New movement, mutation, hybridisation or grouping rules [13–19, 22]	Change what the wolves do each iteration; the candidate pool itself is left unchanged
Improved GWO for engineering design	Nonlinear control parameters, dimension-wise initialization, dynamic position updates [21]	The strongest classical baseline used here; complements rather than duplicates the archive mechanism
Learning-driven and adaptive GWO	Q-learning or reinforcement learning selects strategies, leaders or budgets [23–26]	Report gains against tuned competitors, but rarely against a matched random or fixed-allocation control
Memory, archive and multi-population GWO	Elite memories, cross-search memories, two-archive structures [27, 28]	Closest in spirit; here the archive is *verified* and generates candidates across several regions
GWO for UAV path planning	Q-learning or reinforcement-learning guided trajectory planning [24, 29, 30]	Motivates the 209-dimensional application; the present study adds censored time-to-feasibility and per-map reliability
Surrogate-assisted expensive optimization	Models replace real evaluations and are refined with infill criteria [9–11]	A different regime; here every state variable is backed by a billed real evaluation

missions, time windows. GWO and its hybrids are among the most frequently applied methods in this area [24, 29, 30], alongside genetic and particle-swarm planners and trajectory-optimisation formulations [38–40]. Reported evaluations typically emphasise a single aggregate objective value and a success rate at the final budget, with the budget expressed in iterations. The application study in Section 7 adds three measurements that matter in practice for delivery: the time to first feasible route under a censoring horizon, the cumulative feasibility as a function of the real-evaluation budget, and the final route quality among feasible runs. The three rank methods differently, and separating them allows the trade-offs among the compared algorithms to be described precisely rather than summarised by one number.

2.5 Research gap

Table 1 summarises the comparison set. Operator-level variants change the movement rule; learning-driven variants change the controller; memory mechanisms change what is retained; budget-aware and surrogate methods change how evaluations are modelled or allocated. What has not been examined systematically in GWO is the composition of the candidate pool under a fixed, audited real-evaluation budget—that is, which candidates are *offered* for evaluation when only a small random subset of them can be paid for. This paper addresses that gap, and it does so with a protocol designed to falsify the natural alternatives: identical pool size, identical evaluated subset size, identical budget, matched controls for every scheduling variant, and an out-of-sample seed block for the mechanism study.

3 Proposed method

3.1 Overview and the verified-state discipline

Figure 1 shows one generation of VAST-GWO. A pool of $M = 100$ candidate positions is generated from three routes without any objective evaluation. Exactly $K = 10$ pool members are then sampled uniformly at random without replacement and evaluated with the real objective; each call is billed, and a runtime probe asserts that the audited count equals the budget exactly. The evaluated subset drives all state updates: the leaders α, β, δ, the verified elite archive, and the statistics used by the routes. Unevaluated candidates have

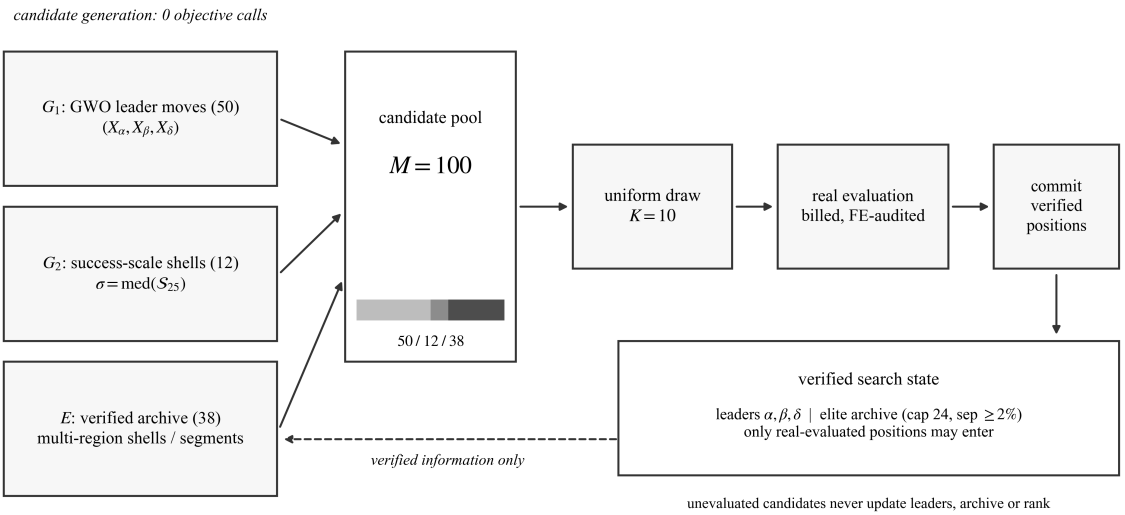


Figure 1. One generation of VAST-GWO. A pool of 100 cheap candidates is generated from three routes; only $K = 10$ uniformly sampled candidates receive a real (billed) evaluation, and only verified information flows back into the leaders and the archive.

no effect on any of them—the verified-state discipline. With $N = 50$ initialization evaluations, a budget of $\mathrm{FE}_{\max} = 100{,}000$ evaluations decomposes into 50 initialization calls plus $(100000 - 50)/10 = 9995$ generations of ten evaluations.

The discipline is not a presentational detail; it is what makes the mechanism auditable. Because no unevaluated position can enter the leaders, the archive or any ranking, every element of the search state is backed by a real measurement, and the improvement of the method can be attributed to a measurable set of decisions. It also removes a failure mode that is common in learned or surrogate-driven variants, where a mechanism can appear to work because a cheap model is optimistic about positions that were never evaluated.

The three routes are complementary by design. Route G1 preserves the classical hunting behaviour and moves the wolves themselves. Route G2 refines the incumbent leader with a step scale that adapts to the geometry of recent successful moves, in the spirit of opposition- and neighbourhood-based refinement [41]. Route E generates candidates from a verified multi-region archive, providing both exploitation of verified basins (single-centre shells) and interpolation between basins (two-centre segments). The allocation of slots across the three routes $(50/12/38)$ and the evaluated subset size $(K = 10)$ were fixed by a pre-registered development scan on development seeds and functions before any formal run was executed (Sections 4–6).

3.2 Candidate generation routes

G1: GWO leader moves (50 candidates). For each wolf i the standard aggregation of the three leader-attracted positions is formed,

$$X_i^{G1} = \frac{1}{3} \sum_{k \in \{\alpha,\beta,\delta\}} \left(X_k - A_k \odot |C_k \odot X_k - X_i| \right), \tag{1}$$

with $A_k = 2a\,r_1 - a$ and $C_k = 2r_2$, where $r_1, r_2 \sim U(0,1)^D$ are drawn per dimension and the convergence factor

$$a = 2 - 2\,\mathrm{FE}/\mathrm{FE}_{\max} \tag{2}$$

is driven by the true evaluation count rather than by an iteration counter. This route preserves the classical exploration–exploitation schedule.

G2: success-scale alpha shells (12 candidates). Local refinement is generated in shells around the incumbent alpha wolf,

$$X^{G2} = X_\alpha + \sigma \odot (2u - 1), \qquad u \sim U(0,1)^D, \tag{3}$$

where the shell scale adapts to the observed geometry of successful moves,

$$\sigma = L\sqrt{3/D}, \qquad L = \max\big(\text{median}(\mathscr{S}_{25}),\ 0.01\, d_{\text{diag}}\big). \tag{4}$$

Here $\mathscr{S}_{25}$ is the set of the 25 most recent *verified* successful step lengths (the Euclidean distance of an accepted move), d_{diag} is the diagonal of the search space, and $\sqrt{3/D}$ converts the per-dimension half-width into an isotropic shell whose expected Euclidean radius equals L. When fewer than five successful steps are available—only at the start of a run—L is initialised to $0.05\, d_{\text{diag}}$. The scale therefore follows the measured geometry of moves that worked instead of a fixed schedule.

E: verified elite archive (38 candidates). The archive stores up to 24 real-evaluated solutions under two rules: a candidate may enter only if it improves the worst archived objective value, and archived solutions are kept separated by at least 2% of d_{diag}. If an evaluated position is not separated from its nearest neighbour but is better than it, the neighbour is replaced instead. Each archived centre carries a *credit* that counts how often it has contributed to an accepted candidate. Nineteen of the 38 archive candidates are single-centre shells. A centre c is drawn with probability proportional to $1 + \text{credit}(c)$, and the candidate is placed on a modulated shell,

$$X^{E_1} = c + \sigma\,(0.5 + v) \odot (2u - 1), \qquad v \sim U(0,1), \tag{5}$$

so the shell radius varies in $[0.5\sigma, 1.5\sigma]$ around the centre. The remaining nineteen are two-centre segments between distinct archived centres c_1, c_2,

$$X^{E_2} = c_1 + \lambda\,(c_2 - c_1) + 0.25\,\sigma \odot (2u - 1), \qquad \lambda \sim U(0.15, 0.85), \tag{6}$$

which interpolate between verified basins. Because archived centres are separated by construction, this route proposes candidates in several distinct regions of the search space at the same time—the property that Section 6 identifies as the source of the method's advantage.

3.3 Sparse allocation and commit rules

The $K = 10$ evaluated candidates are drawn uniformly at random from the pool without replacement. This keeps the allocation free of additional hyperparameters and, as Section 6 shows, is not improved upon by screening, quota or learning-based alternatives. Commits follow the origin of the candidate:

- an evaluated G1 candidate replaces its parent wolf if it improves that parent's objective value;
- an evaluated G2 or E candidate replaces the current worst wolf if it is strictly better than that wolf;
- every accepted move is appended to the verified-step record that drives σ;
- the archive and the leaders α, β, δ are updated exclusively from real-evaluated results.

All candidates are clamped to the search box before evaluation. Clamping is the only boundary handling used; no reflection, re-sampling or repair is applied, so the number of real evaluations consumed per generation is exactly K regardless of the geometry of the search space.

3.4 Archive maintenance

Algorithm 2 states the archive update and credit rule explicitly, because the ablation in Section 6 manipulates this rule. The update is applied to every real-evaluated candidate, including those that do not replace a wolf: the archive therefore accumulates the best verified information of the run rather than only the winners of

Algorithm 1 VAST-GWO (one run)

initialize N wolves by real evaluation; set α, β, δ and $\mathscr{A} \leftarrow \emptyset$; FE $\leftarrow N$
while FE $<$ FE$_{\max}$ **do**
    build pool: 50 G1 moves, 12 G2 shells with scale σ from Eq. (4), 38 archive candidates
    sample $K = 10$ candidates uniformly without replacement
    for each sampled candidate x **do**
        $f(x) \leftarrow$ real evaluation; FE $+= 1$
        commit x according to its route (parent or worst wolf, if better)
        update verified-step record, archive and α, β, δ from verified results only
    end for
end while
return best verified solution

Algorithm 2 Archive update (applied to every verified candidate x, value v)

if $\mathscr{A} = \emptyset$ **then**
    $\mathscr{A} \leftarrow \{x\}$
else if $v <$ max value in $\mathscr{A}$ **then**
    if distance from x to nearest archive centre $\geq 0.02\, d_{\text{diag}}$ **then**
        insert x; if $\mathscr{A}$ is full, replace the entry with the worst value and reset its credit
    else if $v <$ value of the nearest archive centre **then**
        replace that centre by x
    end if
end if
if x came from route E and was accepted, increment the credit of its parent centre

the current generation. When the archive is full, an improving candidate that is sufficiently separated from the stored solutions replaces the worst stored solution; a candidate that is better than its nearest neighbour but too close to it replaces that neighbour. The cold-start case is handled by an explicit fallback: while the archive is empty, the archive slots are filled by α-centred shells, so the pool composition remains constant.

3.5 Complexity

Per generation, VAST-GWO constructs $M = 100$ candidates at cost $O(MD)$, selects $K = 10$ of them, performs K objective evaluations, and maintains the archive at cost $O(|\mathscr{A}|D)$ per evaluated candidate with $|\mathscr{A}| \leq 24$. The arithmetic cost of the search machinery is therefore a few thousand scalar operations per real evaluation at $D = 50$—negligible when evaluations are expensive, and measurable when they are cheap. The number of generations is fixed by the budget at $(\text{FE}_{\max} - N)/K = 9995$, so a run constructs 999,500 candidates for 100,000 evaluations, i.e. ten candidates per evaluation, and the additional arithmetic cost is reported as wall-clock time at equal budget in Section 5.

4 Experimental protocol

4.1 Benchmarks and budgets

All optimization experiments use the CEC2017 real-parameter suite in its official 29-function form (F1 and F3–F30; F2 is omitted as in the official definition) at dimension $D = 50$, with a search domain of $[-100, 100]^D$ and a budget of 100,000 true objective evaluations per run, initialization included [42]. Every algorithm is run 30 times per function on seeds 3001–3030. For a given (function, seed) pair all algorithms start from the same initial population, and the initialization and search random streams are kept separate so that sharing the initial population does not couple the trajectories. Two additional operating points are used to test whether

Table 2. Baselines, sources and configuration under the common protocol ($N = 50$, identical initial populations, 100,000 true evaluations).

Method	Source of the implementation	Configuration under the common protocol
GWO	Original formulation, reimplemented and audited against the published equations [12]	$N = 50$; whole pack evaluated once per iteration; linearly decreasing a; canonical boundary clamping
I-GWO	Published improved GWO [21]	$N = 50$; dimension-wise opposition-based initialization, nonlinear control parameter and dynamic position update as published
HAGWO	Author MATLAB code (commit bdfda96), hierarchical adversarial GWO [46]	$N = 50$; dynamic three-layer division, elite refinement, Lévy perturbation and adversarial escape retained; per-iteration fitness reused instead of re-evaluated
PSO	Classic inertia-weight particle swarm [43, 44]	$N = 50$; w decreasing linearly from 0.9 to 0.4; $c_1 = c_2 = 2$; $v_{\max} = 0.2\,(\mathrm{ub} - \mathrm{lb})$
CLPSO	TEC2006 author code package [45]	$N = 50$; P_c schedule, five-stagnation refresh, $c = 1.49445$; out-of-range positions are not re-evaluated, as in the original code
AOGWO	Author MATLAB code (commit 4122cef), adaptive oppositional GWO [25]	$N = 50$; oppositional initialization, cosine-adaptive a, decaying jump rate with Lévy perturbation and selective opposition retained; search domain adapted to $[-100, 100]^D$

the conclusions are tied to one configuration: $D = 100$ with 100,000 evaluations and ten seeds, and a five-times smaller budget of 20,000 evaluations at $D = 50$ with ten seeds.

The mechanism study in Section 6 uses a disjoint, out-of-sample seed block (3071–3100). All five ablation arms are executed in the same batch with identical pools, identical evaluated subset sizes and identical budgets, so that only the composition of the candidate pool differs.

4.2 Evaluation auditing and parallel execution

Every objective call passes through an independent probe that counts true evaluations, maintains a bounded reservoir of evaluated positions for unbiased novelty estimation, and never perturbs the algorithm's random streams. Each run asserts that the probe count equals the budget exactly, so no method can gain from uncounted evaluations, restarts or cached duplicates. Runs are executed on MATLAB R2025b with exactly four parallel workers (local parpool, no serial fallback); each run writes its result atomically and batches are resumable, so an interrupted batch resumes at the run level without changing the experimental unit. All timings reported below are in-run wall-clock times of the search phase, measured identically for every method on the same machine (8-core AMD Ryzen 7 8745H, 16 threads, 15.3 GB RAM).

4.3 Baselines

Six baselines are compared. Four are established methods: standard GWO [12], the strong improved GWO (I-GWO) with dimension-wise opposition-based initialization, nonlinear control parameter and dynamic position update [21], classic inertia-weight PSO [43, 44] and CLPSO [45]. Two are recent 2026 variants whose authors provide open code: HAGWO, a hierarchical adversarial GWO used here in its author implementation [46], and AOGWO, an adaptive oppositional GWO, also in its author implementation [25]. Table 2 lists the source of each implementation and the configuration used under the common protocol. Each port to the common framework was restricted to interface and accounting levels—initial population supplied by the framework, scheduling variable expressed as a fraction of true evaluations, and one billed evaluation per distinct position—so the search behaviour of every baseline is that of its published or author-released form.

4.4 Statistics

For each baseline we report the mean rank over the 29 functions, the function-level win–loss record, the median paired error ratio of VAST-GWO over the baseline, the pooled p value of the paired Wilcoxon signed-rank test on $\log_{10}$ errors over identical seeds, and the Holm-corrected per-function significant wins and losses. The primary evidence is the per-function paired comparison: for each function the 30 paired runs are tested with a two-sided Wilcoxon signed-rank test on $\log_{10}$ errors [47], and the resulting p values are corrected across the 29 functions with the Holm procedure [48]; the sign of the median paired difference determines whether a significant result counts as a win or a loss. Median paired error ratios below one favour VAST-GWO. Bootstrap 95% confidence intervals for median $\log_{10}$ gains are obtained by resampling the paired differences, and the rank summary is reported alongside Friedman-type ranks over the 29 functions [48, 49]. Parameter settings of every method are the published or author-released ones, not tuned on the test suite [50]. Pooled tests over all function–seed pairs are reported for completeness but are not used as primary evidence, because those pairs are not independent. Effect sizes in the application study are reported as rank-biserial correlations.

Development and formal evaluation were separated. Hyperparameters, slot allocations and the evaluated subset size were selected during a pre-registered development phase on development functions and seeds; the configuration was then frozen, and the formal batches reported here were executed once. All configurations—including those that failed—are retained in the experiment registry. The method's frozen implementation is hash-identified, and a sampling audit confirms bitwise agreement between the frozen configuration and the implementation used in the ablation batches.

5 Results

5.1 Comparison with strong baselines

Table 3 reports the final seven-method comparison and Fig. 2 the corresponding mean ranks. VAST-GWO attains the best mean rank, 2.24, ahead of the strong improved GWO (2.28), the 2026 hierarchical HAGWO (3.41), PSO (4.41), CLPSO (4.48), the 2026 adaptive oppositional AOGWO (5.07) and standard GWO (6.10). Against standard GWO the result is decisive: 28 wins in 29 functions, a median error ratio of 0.525, and 24 significant wins with no significant loss after Holm correction. Against HAGWO, PSO, CLPSO and AOGWO the margins are also significant (median ratios 0.806, 0.652, 0.668 and 0.586; Holm $+19/-3$, $+19/-2$, $+20/-5$, $+21/-0$). I-GWO is the closest competitor. There VAST-GWO leads on the rank summary and on 17 of 29 functions, with a median paired gain of $+0.023$ decades (95% CI $[0.011, 0.032]$); the pooled statistic does not separate the two (ratio 0.938, pooled $p = 0.55$), so the defensible claim is the ranking claim rather than a claim of uniform dominance. Taken together, VAST-GWO is the best-ranked method of the seven and is significantly better than five of six baselines, including both recent GWO variants.

5.2 Where the advantage lies: function families and per-function structure

The aggregate ranking hides a structured picture. VAST-GWO holds the lowest median error of the seven methods on 16 of the 29 functions: both unimodal functions, four of the seven simple-multimodal functions, three of the ten hybrid functions and seven of the ten composition functions. The largest single-function margins are on F9, where the median error is 2.43 against 581.2 for the best competitor (a factor of 238), and on F1, where it is 1.24×10^6 against 7.47×10^6 for the best competitor (a factor of six) and roughly 5×10^3 better than standard GWO. Large but narrower margins occur on F8, F20 and F21 (factors 1.46, 1.39 and 1.16).

Table 4 and Fig. 3 break the comparison down by benchmark family. Against PSO, CLPSO, HAGWO, AOGWO and standard GWO the advantage holds in every family, with ratios between 0.141 and 0.882; the widest margins are on unimodal and simple-multimodal functions and the narrowest on composition functions. Against I-GWO the advantage is family-structured: VAST-GWO leads on the unimodal, simple-

Table 3. Final seven-method comparison (CEC2017, $D = 50$, 100,000 true evaluations, 29 functions, 30 runs on identical seeds). "Median ratio" is the median paired error ratio of VAST-GWO over the baseline (below one favours VAST-GWO); W–L counts functions whose mean error is lower/higher; Holm counts per-function significant wins/losses after Holm correction; time is the mean wall-clock time per run.

Method	Med. ratio	W–L	Holm +/−	Pooled p	Mean rank	Time (s)
VAST-GWO	–	–	–	–	**2.24**	5.93
I-GWO	0.938	17-12	$+12/-6$	0.55	2.28	2.32
HAGWO	0.806	22-7	$+19/-3$	2.2×10^{-36}	3.41	3.41
PSO	0.652	23-6	$+19/-2$	5.8×10^{-31}	4.41	2.59
CLPSO	0.668	21-8	$+20/-5$	7.4×10^{-27}	4.48	2.41
AOGWO	0.586	27-2	$+21/-0$	1.3×10^{-85}	5.07	4.54
GWO	0.525	28-1	$+24/-0$	6.7×10^{-103}	6.10	3.63

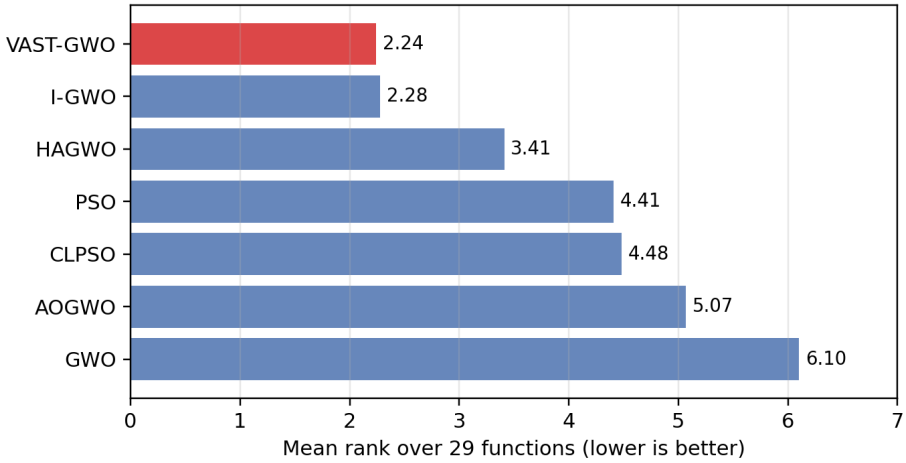


Figure 2. Mean rank over the 29 CEC2017 functions for the seven compared methods ($D = 50$, 100,000 true evaluations, 30 runs each).

Table 4. Function-family median paired error ratios of VAST-GWO over each baseline (below one favours VAST-GWO), with function-level win–loss counts in parentheses. Families follow the CEC2017 classification: 2 unimodal, 7 simple multimodal, 10 hybrid and 10 composition functions.

Baseline	Unimodal	Simple multim.	Hybrid	Composition
I-GWO	0.432 (2-0)	0.796 (4-3)	1.184 (4-6)	0.932 (7-3)
HAGWO	0.227 (2-0)	0.675 (4-3)	0.769 (9-1)	0.882 (7-3)
PSO	0.187 (2-0)	0.557 (5-2)	0.653 (8-2)	0.707 (8-2)
CLPSO	0.144 (2-0)	0.739 (4-3)	0.662 (7-3)	0.701 (8-2)
AOGWO	0.148 (2-0)	0.391 (6-1)	0.513 (10-0)	0.832 (9-1)
GWO	0.141 (2-0)	0.407 (6-1)	0.407 (10-0)	0.733 (10-0)

multimodal and composition families (median ratios 0.432, 0.796 and 0.932) and I-GWO leads on hybrid functions (1.184). The two strongest methods therefore have complementary family profiles, and hybrid functions are the natural target for the next mechanism iteration.

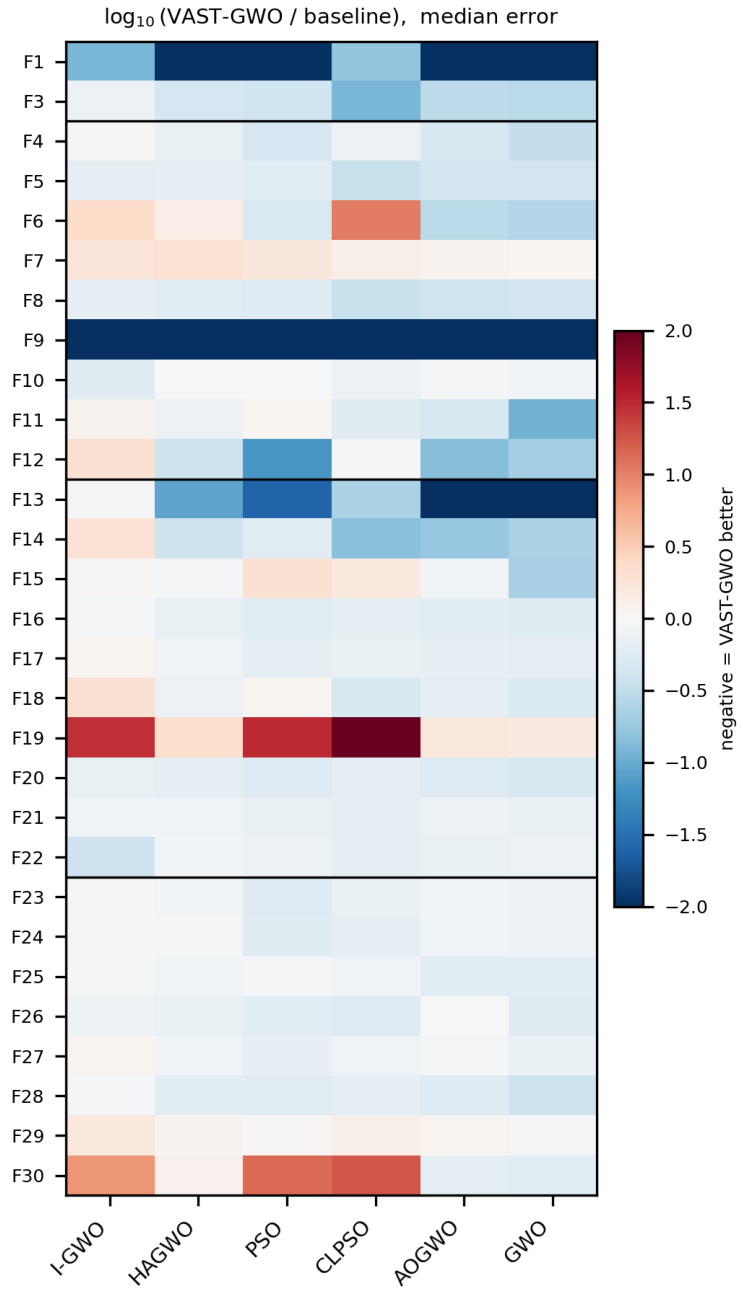


Figure 3. Function-level median error ratios, $\log_{10}(\text{VAST-GWO}/\text{baseline})$, for the six baselines. Blue indicates that VAST-GWO has the lower median error; values are clipped at ± 2 decades. Horizontal lines separate the unimodal, simple-multimodal, hybrid and composition families.

5.3 Convergence and exploration efficiency

The advantage appears early and grows with the budget. At 10%, 20% and 50% of the evaluation budget the median error ratios against standard GWO are 0.802, 0.777 and 0.726, with 25, 26 and 27 of 29 functions respectively better. Against the frozen development-phase reference (ANGWO) the ratios are 0.732 and 0.768 at 10% and 20% of the budget, and on the simple-multimodal subset (F4–F10) the ratio is 0.714. The same trajectories explain how the budget is used: VAST-GWO recovers from stagnation windows more often than standard GWO (recovery probability 0.823 against 0.757) and spends less of the budget inside long stagnation (long-stagnation share 0.813 against 0.844). On the exploration side, the useful-novelty rate—the share of novel evaluations that are followed by a real improvement within a window of $W = \max(50, 1\%\,\text{FE}_{\max})$ evaluations—is 0.00640 against 0.00347 for GWO and 0.00578 for ANGWO, and the

Table 5. Behaviour metrics on the formal tier, computed from the stored best-so-far traces with the definitions given in the text. S1 is the $K = 50$ sparse variant, S3 a fixed quota policy and S5 a Q-learning policy (Section 6); ANGWO is the frozen development-phase reference.

Method	Recovery prob.	Long-stag. share	Useful novelty	Novelty rate Q4	Improving rate Q4
VAST-GWO ($K = 10$)	**0.823**	**0.813**	**0.00640**	0.712	0.00308
S1 ($K = 50$)	0.816	0.812	0.00590	0.695	0.00266
S3 (quota 25%)	0.808	0.821	0.00563	0.745	0.00272
S5 (Q-learning)	0.780	0.835	0.00629	0.716	0.00264
ANGWO (frozen reference)	0.768	0.831	0.00578	0.690	0.00447
GWO	0.757	0.844	0.00347	0.588	0.00481

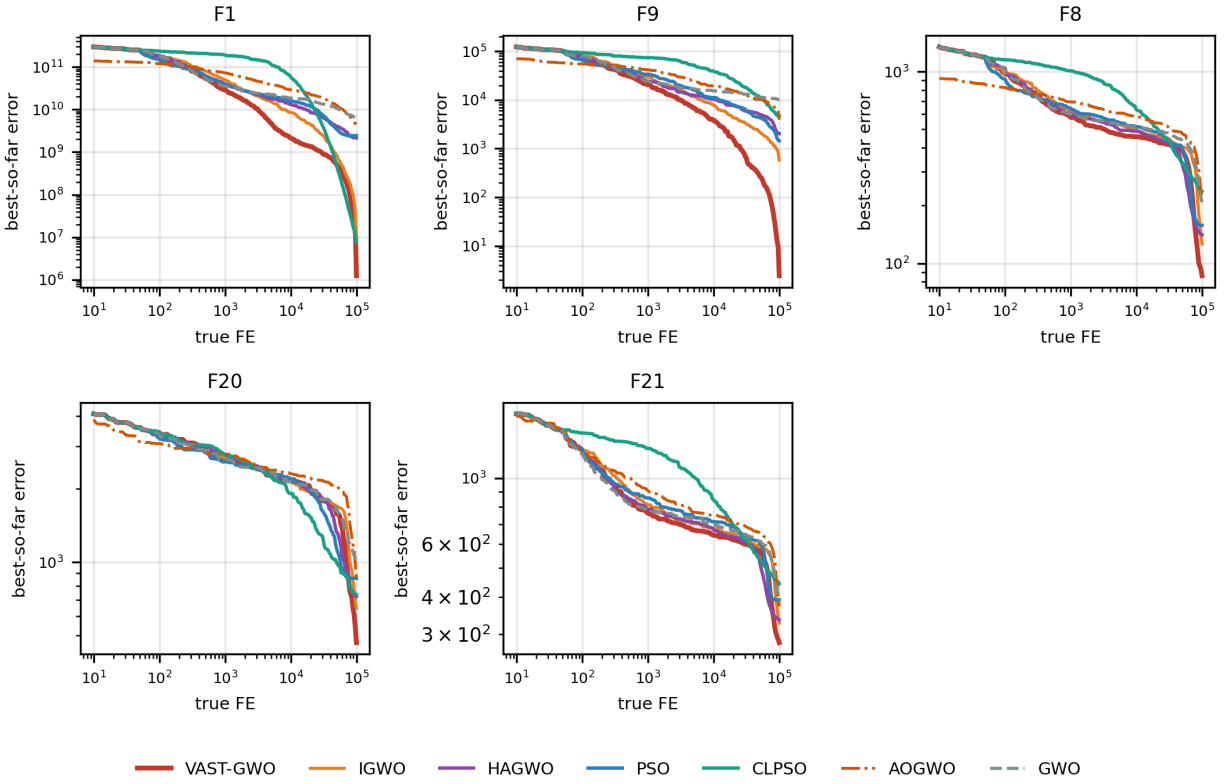


Figure 4. Median best-so-far error of the seven compared methods over 30 runs on five CEC2017 functions. The panels show the five functions on which VAST-GWO attains the largest final margin, one from each benchmark family plus a second simple-multimodal function, so they illustrate the mechanism rather than the average; the complete per-function picture is given by Fig. 3 and Table 3.

novelty rate in the final quarter of the budget is 0.712 against 0.588 for GWO. The improving-evaluation rate in the final quarter is 0.00308 against 0.00481 for GWO, the expected signature of a search that has already located its productive regions and spends the remaining budget opening new ones rather than polishing a single basin. Table 5 reports all behaviour metrics, and Fig. 4 shows representative convergence curves.

5.4 High-dimensional generalization at $D = 100$

A method for expensive black-box optimization earns its place by holding up as the search space grows. We therefore repeat the seven-method comparison at $D = 100$ with the same 29-function CEC2017 suite, the same budget of 100,000 true evaluations, the same population size $N = 50$, and 30 independent seeds (3001–3030). Initial populations are shared per (function, seed) pair and the initialisation and search streams are independent; the protocol, the reuse decision and the statistical plan were frozen before the batch ran. The VAST-GWO runs on seeds 3001–3010 enter the table only after all 290 archived runs were reproduced bit-for-bit (maximum relative difference 0), and the remaining 5,800 runs were executed with exactly four

Table 6. Seven-method comparison on CEC2017 at $D = 100$ with 100,000 true evaluations and 30 seeds. "Ratio" is the median paired error ratio (VAST-GWO divided by the baseline; below one favours VAST-GWO). "W–L" counts functions won and lost on the median error. Holm is the number of functions that remain significant after within-baseline Holm correction. Time is the median of 580 new runs.

Method	Mean rank	Med. ratio	W–L	Holm $(+/-)$	Time (s)	Rel. GWO
VAST-GWO	1.76	–	–	–	8.33	1.98×
IGWO	2.76	0.858	21–8	+14/−6	4.47	1.07×
HAGWO	3.10	0.749	23–6	+19/−3	4.54	1.08×
PSO	4.86	0.598	27–2	+21/−1	3.87	0.92×
CLPSO	4.90	0.565	26–3	+23/−3	4.57	1.09×
AOGWO	5.24	0.484	28–1	+26/−0	6.40	1.52×
GWO	5.38	0.561	27–2	+26/−0	4.20	1.00×

local workers and atomic per-job storage, completing with zero failures.

The advantage does not merely persist at $D = 100$; it grows. VAST-GWO attains a mean rank of 1.76, against 2.76 for I-GWO and 3.10 for HAGWO, improving on its own $D = 50$ rank of 2.24 (Table 6, Fig. 5). Against every baseline the paired function-level Wilcoxon test on $\log_{10}$ errors, Holm-corrected within each 29-test family, returns a decisive net win: GWO $+26/-0$, AOGWO $+26/-0$, CLPSO $+23/-3$, PSO $+21/-1$, HAGWO $+19/-3$ and I-GWO $+14/-6$. The median paired error ratio against I-GWO tightens from 0.938 at $D = 50$ to 0.858, and against HAGWO from 0.806 to 0.749; a global Holm correction over all $6 \times 29 = 174$ tests leaves every direction unchanged. Fig. 6 shows the same behaviour at the level of individual trajectories: VAST-GWO separates from the pack early and keeps opening ground through the last checkpoint, which is what a sparse-evaluation design should deliver when the search space expands.

The margin is concentrated where the mechanism is designed to act. Across the six baselines the median error ratio is 0.41 on unimodal and 0.44 on hybrid functions, against 0.71 on the composition class, where all methods converge towards comparable basins and the archive's multi-region memory buys less. This is the expected operating envelope of archive-guided sparse evaluation: its value rises with the number of competing basins a fixed evaluation budget must cover. Within the strongest comparison, I-GWO remains the closest competitor: it resolves six functions in its own favour (F6, F15, F19, F27, F29 and F30, predominantly composition and hybrid cases), while VAST-GWO resolves fourteen in its favour and leaves nine statistically tied. The per-function resolution is reported in full in the supplementary per-function table and in Fig. 7, so that the boundary of the envelope is documented rather than inferred.

The comparison is conducted at equal true-evaluation cost, and VAST-GWO pays for its candidate pool in wall-clock time: 8.33 s per run at $D = 100$, 1.98× GWO (Fig. 8). This is the arithmetic cost of constructing 100 candidates per generation and is the price of the search structure, not of any additional objective call: every run is asserted at exactly 100,000 real evaluations by an independent probe. Because the methods differ in how many evaluations a generation consumes, equal FE is not equal iteration count—VAST-GWO performs about 9,995 generations, GWO, HAGWO and PSO about 1,999, and I-GWO about 1,000—so the reported gains are gains at equal evaluation cost, the resource that dominates in the expensive-optimization regime this paper targets.

5.5 Additional operating points

Table 7 reports the small-budget operating point. With a budget five times smaller, VAST-GWO ranks first (1.97), winning 24 of 29 functions against standard GWO (ratio 0.721), 24 of 29 against the frozen reference (0.807) and 24 of 29 against its dense variant (0.678). The sparse-evaluation candidate pool therefore scales upward in dimension and downward in budget, which is the regime relevant to engineering problems where large budgets are unavailable.

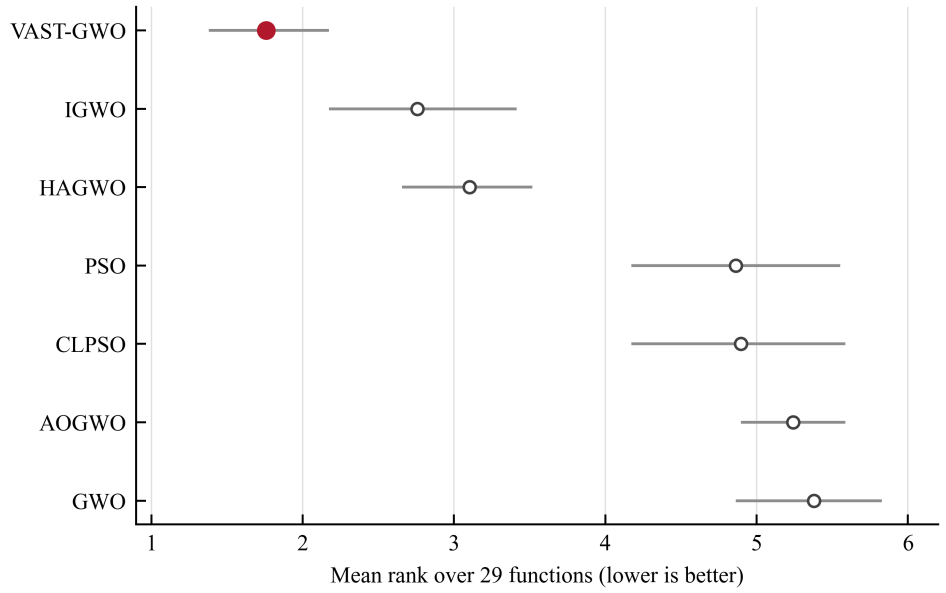


Figure 5. Mean rank over the 29 CEC2017 functions at $D = 100$ (lower is better). Error bars are 95% function-level bootstrap intervals.

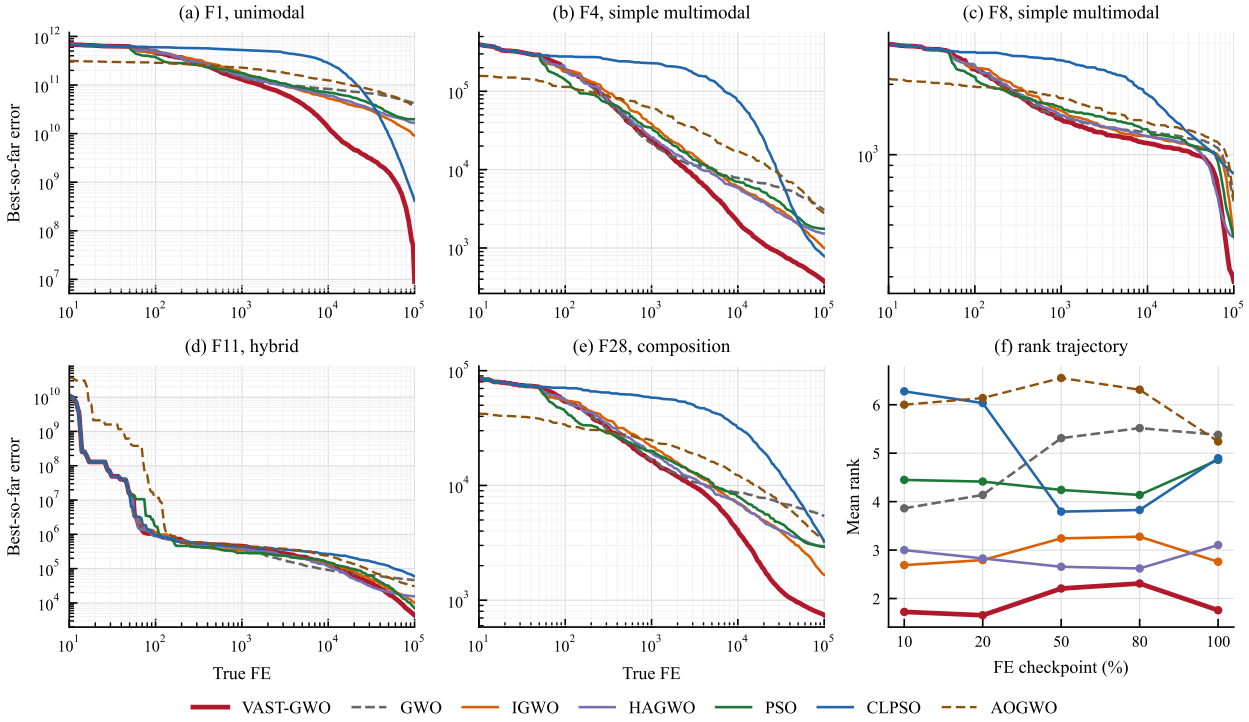


Figure 6. Convergence at $D = 100$ (median of 30 runs). Panels (a)–(e) show the best-so-far error against true function evaluations on one representative function per benchmark family plus a second simple-multimodal case; panel (f) gives the mean rank at 10, 20, 50, 80 and 100% of the budget. The rank advantage of VAST-GWO is established at the first checkpoint and holds to the full budget.

5.6 Computational cost

At equal budgets VAST-GWO is the most expensive method in wall-clock terms: 5.93 s per run at $D = 50$ with 100,000 evaluations, against 3.63 s for GWO, 3.41 s for HAGWO, 2.59 s for PSO, 2.41 s for CLPSO, 2.32 s for I-GWO and 4.54 s for AOGWO; at $D = 100$ the same budget costs 8.33 s against 4.20 s for GWO, 4.47 s for I-GWO and 6.40 s for AOGWO (Table 6). The overhead relative to GWO is 1.6× at $D = 50$ and 1.98× at $D = 100$, and originates in pool construction and archive maintenance, both of which are arithmetic.

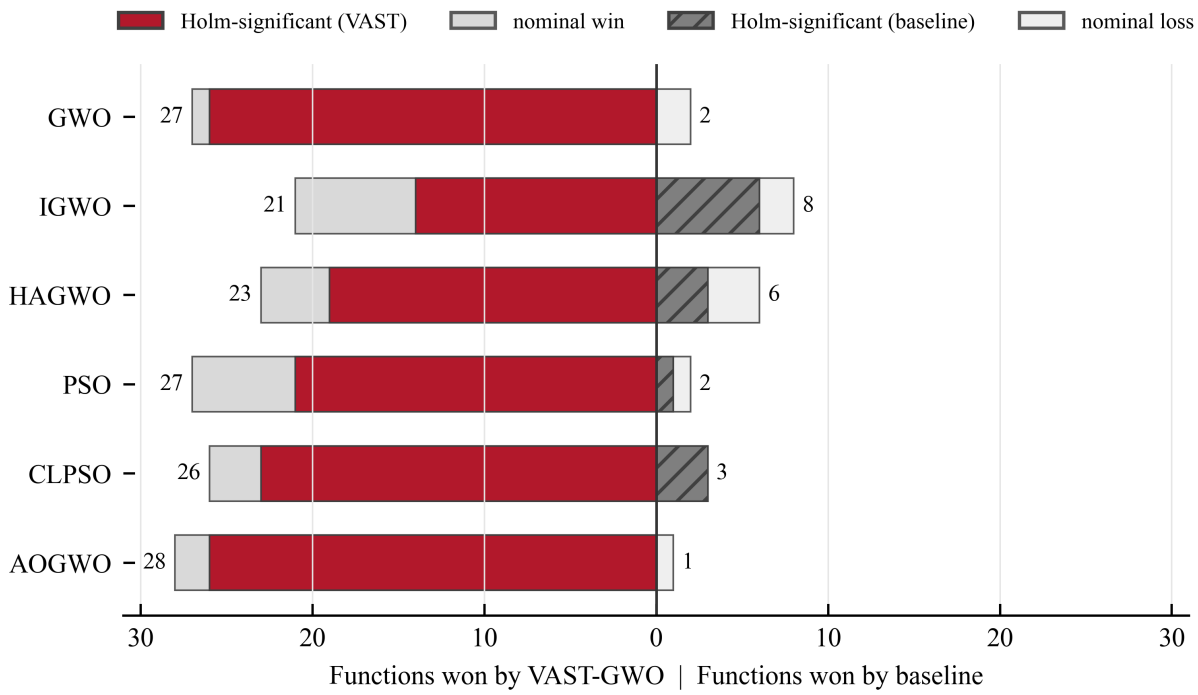


Figure 7. Per-function win/loss counts against VAST-GWO at $D = 100$, and the subset that remains significant after within-baseline Holm correction.

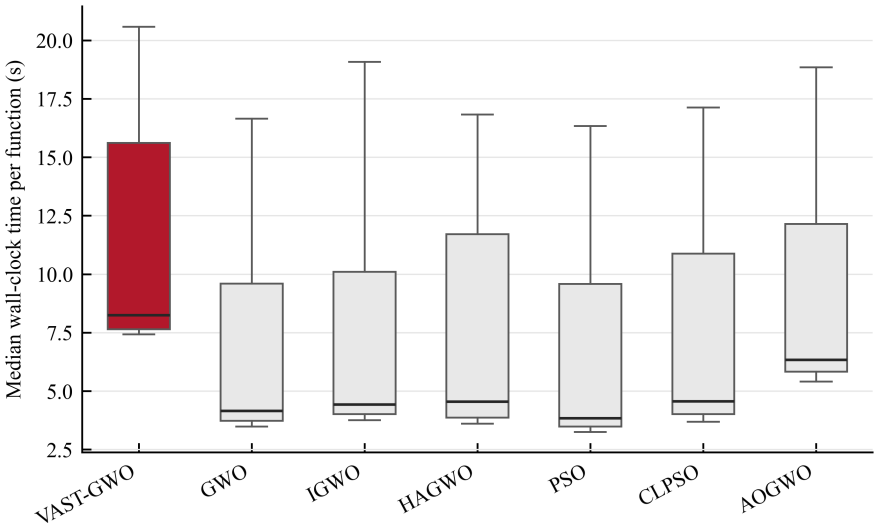


Figure 8. Wall-clock time at $D = 100$ (median per function over seeds 3011–3030, 100,000 true evaluations per run).

In the regime VAST-GWO targets—a scarce number of expensive evaluations—this cost is negligible compared with the objective, and it buys both the leading rank and the stagnation recovery reported above. In the opposite regime, where evaluations are nearly free, the runtime figures above allow the trade-off to be evaluated directly rather than assumed.

6 Ablation and mechanism analysis

6.1 Equal-slot source ablation

The ablation answers a single question by design: with the pool size, the evaluated subset size, the budget and the acceptance rules held constant, which compositions of the candidate pool produce the performance?

Table 7. Small-budget configuration: 20,000 evaluations, $D = 50$, 29 functions, ten seeds. Ratios are relative to VAST-GWO (below one favours VAST-GWO) with win–loss counts in parentheses.

Method	Mean rank	Med. ratio vs VAST-GWO
(b) 20k FE, $D = 50$, 29 functions, 10 seeds		
VAST-GWO	**1.97**	–
S1-K50	2.31	0.979 (17-12)
ANGWO	3.10	0.807 (24-5)
B1-dense	3.79	0.678 (24-5)
GWO	3.83	0.721 (24-5)

Table 8. Equal-slot source ablation. All arms use $M = 100$, $K = 10$, 100,000 true evaluations and identical acceptance rules; only the composition of the pool differs.

Arm	Candidate pool (equal total $M = 100$)	Mechanism under test	Rationale
A0	100 GWO leader moves	Sparse evaluation alone	Isolates the effect of evaluating only K of M candidates
A1	50 GWO moves + 50 α-shells	Leader-centred refinement	Tests whether local refinement around α can replace multi-region memory
A2	50 GWO moves + 50 archive candidates	Verified multi-region archive	Isolates the archive without the α-shell route
A3	50 GWO moves + 12 α-shells + 38 archive candidates	Full VAST-GWO	Reference configuration
A5	50 GWO moves + 12 α-shells + 38 single-centre draws around α	Archive replaced by a single centre with identical slots, scale and acceptance rule	Tests whether several regions, rather than extra perturbation, produce the gain

Table 8 lists the five arms. Arm A0 contains only standard GWO leader moves and therefore measures what sparse evaluation alone achieves. A1 adds leader-centred refinement, A2 adds the verified archive, A3 is the full method, and A5 replaces the archive by a single alpha-centred sampler with identical slot count, identical shell scale and identical acceptance rule, which isolates the multi-region property of the archive from the mere presence of extra perturbation.

Table 9 reports the outcome. Removing both auxiliary routes (A0) degrades the method to 27 losses in 29 functions (median ratio 0.538, Holm $+23/-0$): sparse evaluation by itself is not the source of the advantage. Removing the archive while keeping the alpha shells (A1) still leaves the method 25 functions behind the full version (0.812, Holm $+21/-1$). Keeping the archive but removing the alpha shells (A2) leaves performance statistically unchanged (0.994, 14–15, pooled $p = 0.326$), so the alpha-shell route is optional once the archive is present. Finally, replacing the archive by a single-centre sampler with identical slots, scale and acceptance rule (A5) collapses the method to 2 wins in 29 functions (1.407, Holm $+2/-23$). The mean ranks over the five arms mirror the same ordering: A3 1.72, A2 1.79, A5 3.34, A1 3.52, A0 4.62.

Two conclusions follow. First, the verified elite archive is the performance engine of VAST-GWO: A2, which contains only GWO moves and archive candidates, is statistically indistinguishable from the full method while beating the GWO-only pool 29–0. Second, it is the *multi-region* character of the archive that matters rather than the amount of additional perturbation: A5 uses the same number of extra candidates,

Table 9. Equal-slot candidate-source ablation (29 functions, 30 seeds 3071–3100, 100,000 true evaluations each). Ratios are relative to the full configuration; Holm counts the significant wins/losses of the listed comparison.

Comparison	Median ratio	W–L	Holm $+/-$	Pooled p
A3 (full) vs A0 (GWO moves only)	0.538	27–2	$+23/-0$	7×10^{-102}
A3 (full) vs A1 (G1+G2)	0.812	25–4	$+21/-1$	5×10^{-60}
A3 (full) vs A2 (G1+E)	0.994	14–15	$+1/-0$	0.326
A2 (G1+E) vs A0	0.510	29–0	$+27/-0$	7×10^{-117}
A5 (single centre) vs A3 (full)	1.407	2–27	$+2/-23$	1×10^{-64}

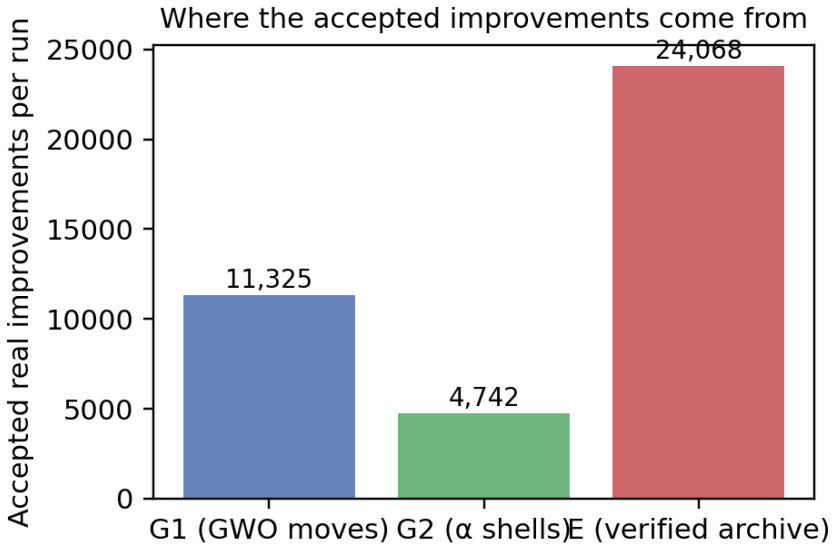


Figure 9. Accepted real improvements per run by candidate route, averaged over the ablation batch. The verified archive route dominates; the alpha-shell route is the smallest contributor, consistent with its optional role once the archive is present.

the same shell scale and the same acceptance rule, and differs from the full method only in drawing those candidates from one centre instead of several, yet its performance falls by a factor of 1.41. The source-level attribution in Fig. 9 confirms the same picture at the level of accepted events: in a full run the archive route produces on average 24,068 accepted improvements, against 11,325 for the GWO-move route and 4,742 for the alpha-shell route.

6.2 Design-space boundaries and rejected alternatives

The same controlled protocol was used to test the natural alternatives to uniform random allocation and to two popular enhancement strategies (Table 10). Every one of them was rejected by its pre-registered criterion. Directed screening by proximity to the incumbent and by archive credit is worse than uniform sampling (median ratio 1.394) because it starves the GWO route of evaluations; informed stratified allocation (1.528), pool-ratio allocation (1.255) and trend-informed allocation (1.348) are also worse than the uniform draw; fixed exploration quotas of 0.25, 0.50 and 0.75 degrade performance (1.070, 1.064 and 1.672); and three Q-learning quota schedulers—with zero, tuned and prior-initialised Q tables—never exceed the best fixed policy (0.988, 1.024 and 1.130), although all of them beat random quota perturbation (0.748–0.870). A stagnation-triggered radius boost degrades final quality (1.204). Finally, steering moves along previously successful directions improves the final error against the frozen configuration (0.897, 21 wins in 29 functions), but an equal-magnitude random-direction control reproduces the effect (0.988 between the two steering variants, pooled $p = 0.155$), and the gain appears only in the last 20% of the budget while wall-clock time rises by

Table 10. Design-space boundary experiments. Each hypothesis was tested with a matched control under the common protocol; ratios below one favour the control variant and ratios above one favour the reference configuration.

Hypothesis	Control	Median ratio	Verdict
Screening the best-looking candidates beats random allocation	Directed screening by proximity to α and archive credit (S2)	1.394 vs uniform	Rejected; it starves the GWO route of evaluations
Stratified or informed allocation beats a uniform draw	Informed-in-strata, pool-ratio and trend-informed allocation	1.528 / 1.255 / 1.348	Rejected
A fixed exploration quota manages the budget better than the pool share	Fixed quota at 0.25 / 0.50 / 0.75	1.070 / 1.064 / 1.672	Rejected; raising the quota above the pool share is harmful
Reinforcement learning schedules the budget better than the best fixed policy	Three Q-learning quota schedulers (zero, tuned and prior-initialised)	0.988–1.130 vs best fixed; 0.748–0.870 vs random quota	Rejected as a contribution; always at or below the best fixed policy
A stagnation-triggered radius boost restores progress	Shell radius multiplied by three after stagnation	1.204	Rejected
Reusing verified successful directions adds information beyond extra perturbation	Steering along the mean recent successful direction	0.897 vs frozen; 0.988 vs equal-magnitude random directions ($p = 0.155$)	Rejected; the random control reproduces the gain

17–25%. The mechanism was therefore rejected as an information-driven contribution and is not part of VAST-GWO.

6.3 Interpretation of the mechanism evidence

Three statements are supported by the ablation and boundary results, and one is not. Supported: (i) sparse evaluation alone is not sufficient—the GWO-only pool loses 27 of 29 functions to the full method; (ii) verified multi-region memory is the dominant source of the improvement—removing the archive costs 25 functions, replacing it with a single centre costs 27, and the archive route accounts for the majority of accepted improvements; (iii) the allocation rule does not need to be learned or tuned—uniform sampling dominates or matches every alternative tested, including three Q-learning schedulers with matched controls. Not supported: alpha-centred local refinement is not necessary once the archive exists, since A2 and A3 are statistically indistinguishable. The implication is a design rule rather than a mechanism stack: make the candidate pool contain verified information from several regions, and keep the evaluation rule simple.

7 Application: 3D UAV delivery path planning

7.1 Problem formulation

The application is a three-dimensional UAV delivery problem with hard time windows, built on real terrain data. Each instance contains 20 delivery stations with individual time windows, a depot, three cylindrical threat zones and three intermediate control points per flight leg. A candidate solution is a decision vector of dimension

$$D = 20 + \underbrace{21 \times 3 \times 3}_{\text{21 legs, 3 control points, 3 offsets}} = 209. \tag{7}$$

The first 20 coordinates define the visiting order through their rank; the remaining 189 coordinates define, for each leg, three control points by a position ratio along the straight line from the current point to the next target, a lateral offset and a height offset. The decoded trajectory is a sequence of legs from the depot through all stations and back to the depot.

Table 11. Parameters of the 3D UAV delivery benchmark (benchmark-v1). The benchmark is a normalized simulation derived from real terrain data, not a flight test.

Quantity	Value	Role in the model
Decision vector	209 dimensions	20 ranking values for the visiting order + 189 control offsets (21 legs $\times$ 3 control points $\times$ 3 offsets)
Map extent	100×100 units	Horizontal extent of each terrain tile; depot at $(40, 50)$
Terrain	Copernicus GLO-90, five tiles	Ridge, peaks, mountain, hills and plateau; bilinear interpolation on a 51×51 grid
Delivery points	20 per map	Fixed station coordinates; five-digit hard time windows per station
Cruise speed	10 units per time unit	Converts leg length to arrival time
Service height	16 m above ground	Station altitude; depot altitude follows the terrain
Terrain clearance	4 m	Minimum vertical separation between route and terrain along each leg
Threat zones	3 cylinders per map	Radius plus 2 m safety margin; vertical extent from terrain base to a specified ceiling
Operating ceiling	42 m	Maximum flight altitude
Climb / turn limits	35° / 120°	Maximum flight-path angle and heading change within a leg
Objective weights	1 / 0.05 / 2	Distance, waiting time and smoothness terms in Eq. (8)
Penalties	$10^5 + 10^3 \Delta t$ / $10^5 + 10^4 \Delta g$	Time-window lateness and geometric violations, respectively

The objective combines total flight distance, waiting time and path smoothness,

$$J = \sum_{\ell} d_{\ell} + 0.05 \sum_{\ell} w_{\ell} + 2 \sum_{\ell} s_{\ell}, \tag{8}$$

where d_{ℓ} is the length of leg ℓ, w_{ℓ} is the waiting time accumulated before a station's ready time, and s_{ℓ} penalises turn and pitch changes. Hard constraints are imposed on terrain clearance, threat-zone avoidance, maximum altitude, climb and turn limits, the delivery time windows and the return to the depot, with penalty terms that keep infeasible solutions strictly behind feasible ones. Table 11 lists the model parameters; the five scenarios use real Copernicus GLO-90 terrain tiles with distinct morphology (ridge, peaks, mountain, hills and plateau) and were frozen before the formal runs.

7.2 Protocol and metrics

All six methods run with $N = 50$ and 30,000 true evaluations per run, initialization included, on 30 repetitions for each of the five scenarios, giving $6 \times 5 \times 30 = 900$ independent runs. Each method starts from the same initial population for a given (map, repetition), the search random stream is offset identically, and every run is verified after the search with the full evaluator outside the billed budget. Four quantities are reported because they answer different engineering questions: the final feasibility rate; the time to the first feasible route, summarised by the restricted mean survival time (RMST) under a 30,000-evaluation censoring horizon, in which every run enters the statistic; the cumulative feasibility at 2k, 5k, 10k, 20k and 30k evaluations; and the final route cost on paired runs in which both methods were feasible. Final feasibility is compared with McNemar tests, route cost with paired Wilcoxon tests and rank-biserial effect sizes, and time to feasibility with a censoring-aware ordered test on the checkpoint at which feasibility was first reached.

7.3 Results

VAST-GWO is the only method that finds a feasible route in every one of its 150 runs; the runner-up reliability is 97.3% (I-GWO and HAGWO), followed by PSO (95.3%), GWO (94.0%) and CLPSO (61.3%). It also at-

Table 12. 3D UAV delivery application (150 runs per method: five maps, 30 repetitions). Feasibility is the share of runs that found a feasible route within 30,000 evaluations; RMST is the restricted mean time to first feasibility under a 30,000-evaluation censoring horizon, so all runs enter the statistic; the first-feasible median is conditional on successful runs and is shown for reference; cost is the median final route cost over feasible runs (lower is better).

Method	Feasible (%)	RMST	RMST 95% CI	First feasible[†]	Cost[‡]
VAST-GWO	**100.0**	**7062**	[6449, 7731]	6314	667.4
GWO	94.0	10936	[9941, 11983]	8920	699.8
I-GWO	97.3	12270	[11374, 13179]	11152	724.7
HAGWO	97.3	8133	[7355, 8950]	7432	663.4
PSO	95.3	8138	[7058, 9324]	5357	766.4
CLPSO	61.3	24427	[23312, 25383]	21330	1015.4

Table 13. Per-map feasibility (% of 30 runs) on the five terrain scenarios.

Method	Hills	Mountain	Peaks	Plateau	Ridge
VAST-GWO	**100**	**100**	**100**	**100**	**100**
GWO	96.7	83.3	100	90.0	100
I-GWO	100	86.7	100	100	100
HAGWO	96.7	93.3	100	96.7	100
PSO	100	83.3	100	96.7	96.7
CLPSO	56.7	36.7	80.0	60.0	73.3

tains the smallest RMST, 7,062 evaluations (95% CI 6,449–7,731), ahead of HAGWO (8,133), PSO (8,138), GWO (10,936), I-GWO (12,270) and CLPSO (24,427). The cumulative feasibility curves show where the advantage appears: at 10,000 evaluations VAST-GWO has already reached 82.0% feasibility, against 76.0% for HAGWO and 74.0% for PSO, and at 20,000 evaluations it reaches 100% while every other method is still between 26.7% and 97.3%. Table 12 summarises reliability and quality, Figs. 10–13 show the behaviour in detail, and Table 13 reports the per-map feasibility rates, where VAST-GWO achieves 100% on all five terrains.

Final route quality favours VAST-GWO against every baseline. Over the pairs in which both methods were feasible, the median paired cost ratios are 0.939 against GWO, 0.898 against I-GWO, 0.967 against HAGWO, 0.843 against PSO and 0.631 against CLPSO, and all five remain significant after Holm correction (p from 3.3×10^{-16} to 0.0458, rank-biserial 0.19–1.00). The censoring-aware ordered test is significant against GWO, I-GWO, HAGWO and CLPSO; the exception is PSO ($p = 0.598$), which finds feasible routes earlier in the first few thousand evaluations (median 5,357 against 6,314 in successful runs) and then stalls: its final routes are 15% longer (median cost 766.4 against 667.4) and it misses feasibility in seven runs. HAGWO's marginal median cost (663.4) is slightly below that of VAST-GWO (667.4) when each method's own successful runs are considered separately; on the 146 runs where both methods succeed, VAST-GWO is significantly better (ratio 0.967, Holm $p = 0.0458$). McNemar tests on final feasibility are significant against GWO ($p = 0.016$), PSO ($p = 0.047$) and CLPSO ($p < 10^{-16}$) after Holm correction, while the reliability difference against I-GWO and HAGWO—both of which fail four runs—is not significant on four discordant pairs.

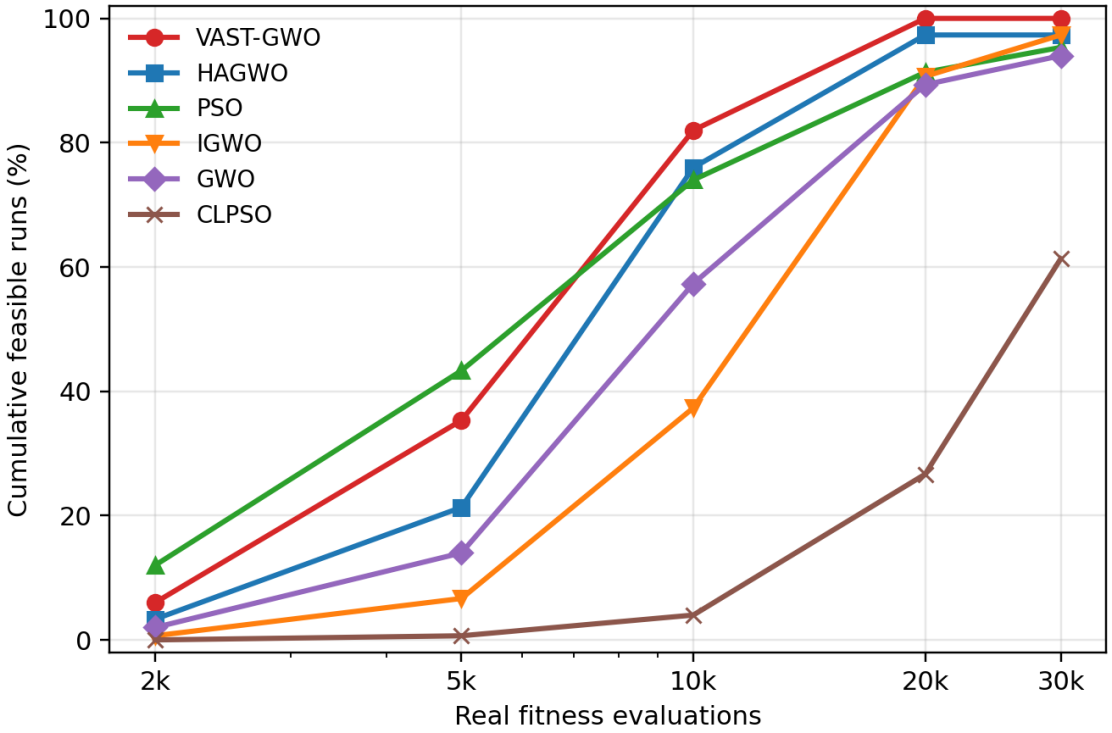


Figure 10. Cumulative feasible runs versus real evaluations on the five DEM-based delivery scenarios (150 runs per method). Checkpoints are early observations within the single 30k protocol.

7.4 Discussion of the application results

The application reproduces the mechanism in a setting where the objective is 209-dimensional, the constraints are hard and the terrain is real: the method that combines sparse evaluation with a verified multi-region archive is the only one that is reliable on all five maps, it reaches feasibility faster than every method except PSO—and that exception reverses by the end of the budget—and it produces better final routes than all five baselines on the paired runs where the comparison is meaningful. The trade-off that PSO exhibits is instructive for practice: reaching a feasible route slightly earlier is attractive in operations, while the final route quality and the seven infeasible runs are what a delivery plan is ultimately judged on. The UAV results are therefore not an additional leaderboard but an independent demonstration of the paper's central claim: with a scarce real-evaluation budget, the composition of the candidate pool determines both whether feasible solutions are found at all and how good they become.

8 Discussion

8.1 Candidate-source structure is a first-class design variable

The experiments support a compact statement about optimizer design under scarce real evaluations: *the composition of the candidate pool is a first-class design variable, and engineering it can outweigh engineering the evaluation scheduler*. In VAST-GWO that composition is a verified archive spanning several regions of the search space. Because every archived centre is backed by a billed evaluation, the archive can safely generate candidates far from the current leaders without introducing unverified optimism; because it spans multiple regions, it counteracts the leader-driven contraction that makes GWO stagnate on multimodal landscapes; and because the allocation remains a simple uniform draw, none of the gain depends on a fragile controller. The boundary experiments make the contrast explicit: of the scheduling mechanisms tested—directed screening, informed stratification, fixed quotas, random quotas and three Q-learning policies—none

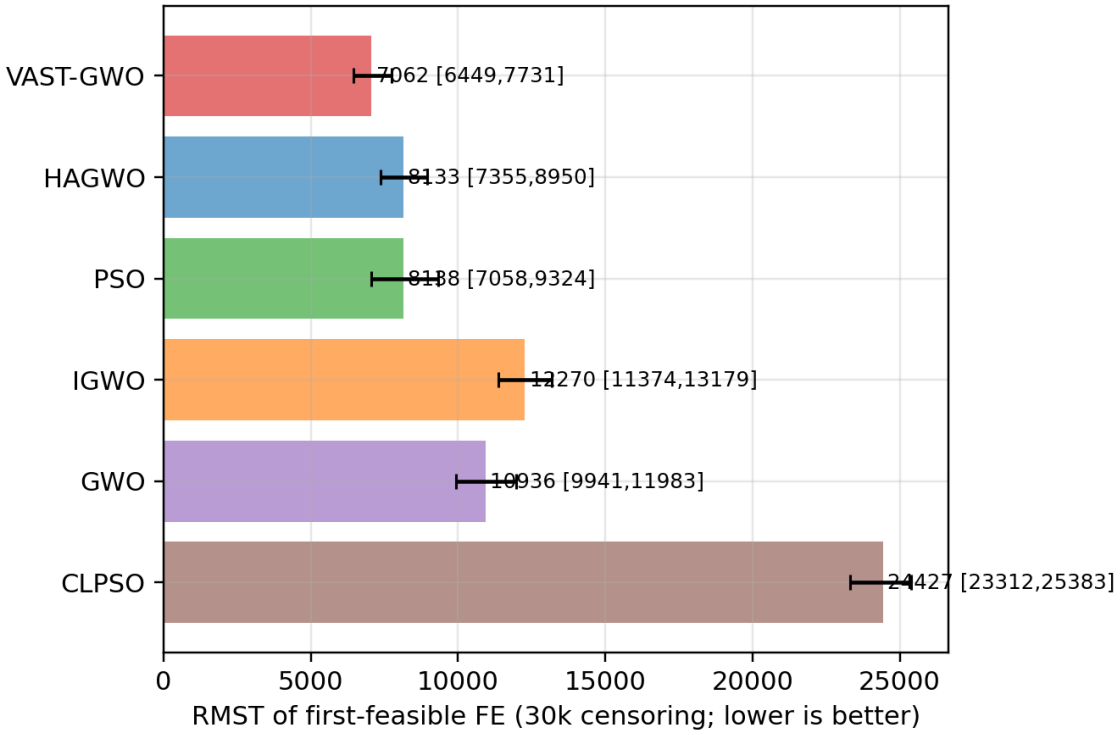


Figure 11. Censored mean time to first feasible route (RMST, 30k horizon, bootstrap 95% confidence intervals). Every run enters the statistic.

improved on the uniform draw, and the learned policies were never better than the best fixed policy. The boundary evidence is what makes the design rule actionable: it shows that the gain comes from the information the pool contains, not from a particular way of spending the budget, so the mechanism can be adopted without importing a tuning or learning pipeline.

The behaviour metrics describe how the mechanism produces its effect. VAST-GWO recovers from stagnation windows more frequently than standard GWO (0.823 against 0.757) and spends less of the budget inside long stagnation (0.813 against 0.844). At the same time, its evaluations are individually more valuable: the useful-novelty rate is almost twice that of standard GWO, and the novelty rate in the final quarter remains high, which indicates that the method continues to open new ground late in the run instead of refining a single basin. These two observations correspond exactly to the dual objective of the study—stagnation recovery and exploration efficiency under a fixed true-evaluation budget—and both are measured from the same audit trail as the final errors.

8.2 Why the verified-state discipline matters

Three of this paper's claims depend on the counting discipline rather than on the algorithm alone. The comparison with strong baselines is meaningful only if every method consumes exactly the same number of real evaluations, which is why an independent probe audits every call. The ablation is meaningful only if the evaluation budget, the pool size and the accepted-subset size are held constant, which is why the disabled routes in the ablation are replaced by standard candidates rather than removed. And the mechanism claim is meaningful only if the archive contains measured information, which is why no unevaluated candidate is allowed to influence leaders, archive or rankings. The same discipline makes the method reproducible: the frozen configuration is hash-identified, every configuration tested during development is retained (including the ones that failed), and the formal batches are executed once after freezing.

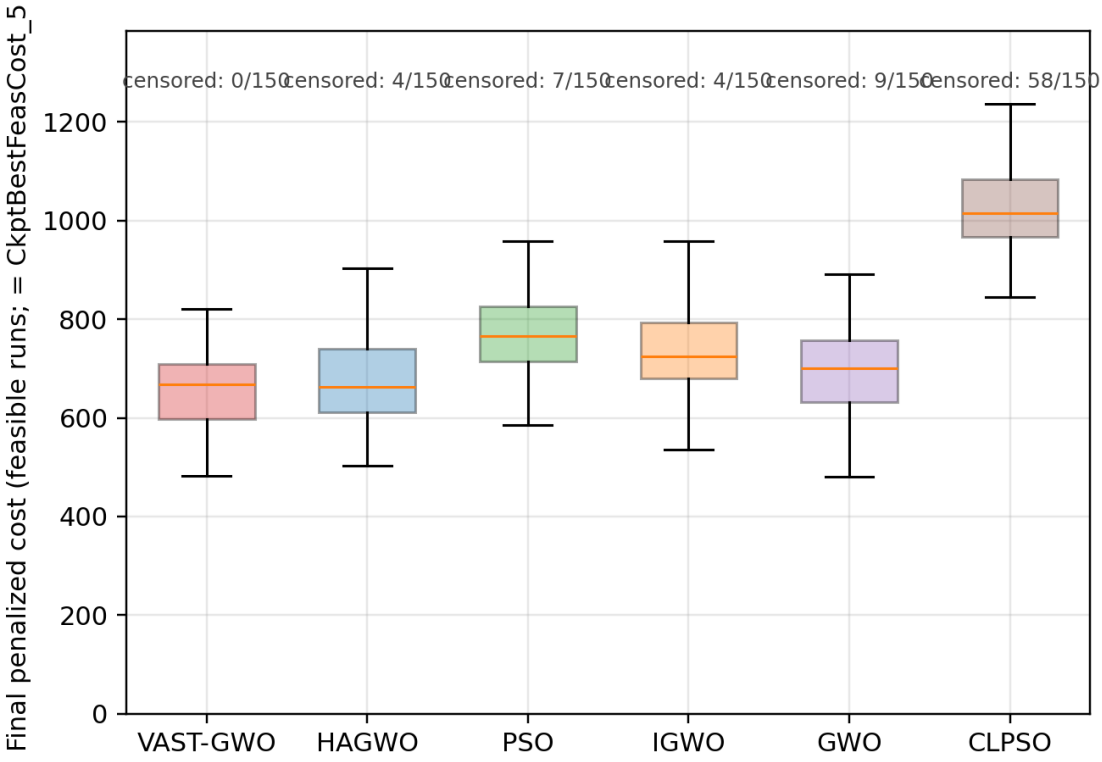


Figure 12. Distribution of final feasible route cost per method over the five scenarios and 30 repetitions.

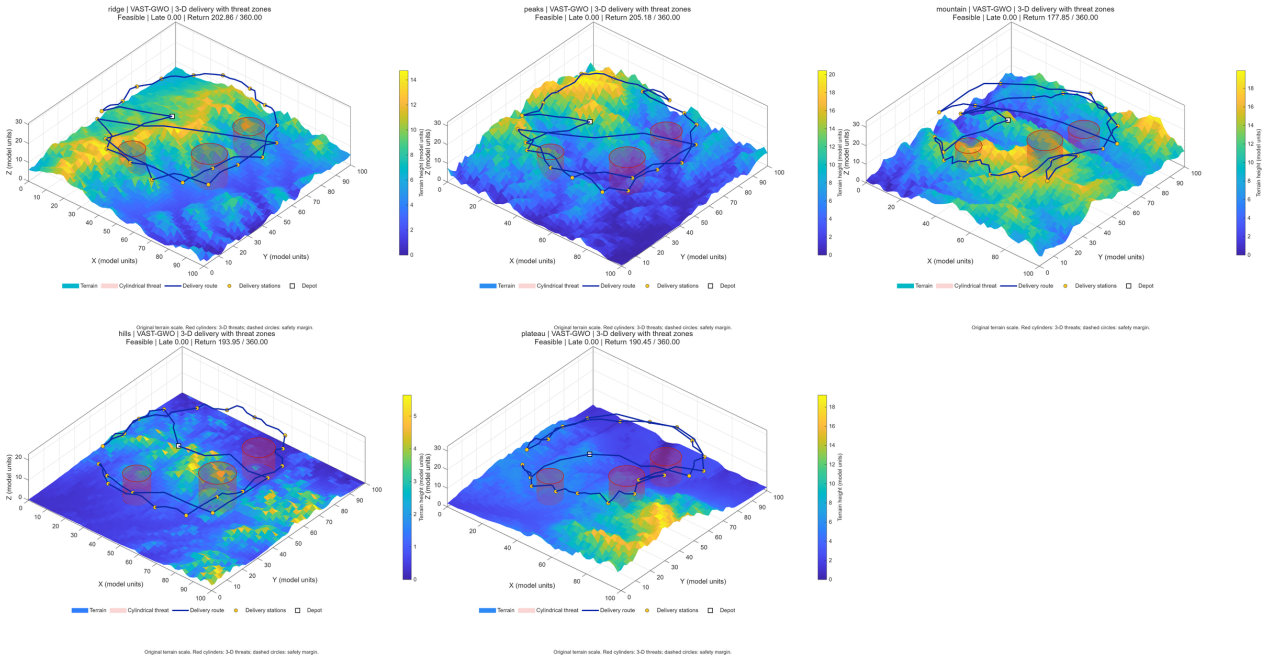


Figure 13. Best feasible delivery routes found by VAST-GWO on the five DEM-based scenarios (costs 481.7–600.7). Terrain colour encodes elevation; red cylinders are threat zones with safety margins; the blue polyline is the route, dots are stations and the square is the depot.

8.3 Relation to the strong improved GWO

The comparison with I-GWO is the informative one. VAST-GWO is ranked first of the seven methods and is significantly better than five of the six baselines, including both 2026 variants; I-GWO is the single baseline that is not separated on the pooled statistic, and its strength is concentrated in the hybrid family. At $D = 100$ the ordering widens in favour of VAST-GWO (rank 1.76 against 2.76; median ratio 0.858). The

complementary family profile points to a concrete next step: hybrid functions combine subcomponents with different landscapes, which is exactly where archive candidates can be complemented by a stronger local search inside each subcomponent.

8.4 Scope of the evidence

The claim of this paper is deliberately bounded, and each boundary is a design choice rather than an open question about the mechanism. The benchmark evidence covers CEC2017 at $D = 50$ and $D = 100$ with budgets from 20,000 to 100,000 true evaluations, which is the regime in which the evaluation count, not the arithmetic count, is the binding constraint. The application evidence covers one family of terrain-constrained delivery problems with hard time windows, simulated on five real terrain tiles and therefore an engineering-grade testbed rather than a flight test. The archive parameters (capacity 24, separation threshold 2% of the diagonal) are fixed constants of the method rather than per-problem tuning knobs, and they were held constant across both dimensions and both budget levels reported here, so the same configuration transfers without re-tuning. Wall-clock cost is the one resource VAST-GWO spends deliberately: it is $1.6\times$ standard GWO at $D = 50$ and $1.98\times$ at $D = 100$, which is the correct trade when each objective call is expensive and the wrong one when evaluations are free—a boundary the runtime figures make explicit so that a practitioner can decide directly.

8.5 Future work

Three directions follow directly from the results. First, the ablation identifies a simpler configuration: because A2 is statistically indistinguishable from the full method, a two-route variant that keeps only the leader moves and the archive deserves a dedicated study, both for simplicity and to test whether the archive alone can carry even more of the budget. Second, the verified archive is a natural training set for a lightweight surrogate, because every entry is a real measurement; coupling the two would extend the method to objectives whose evaluations are expensive by orders of magnitude. Third, the application evidence motivates delivery problems with moving obstacles, stochastic time windows and multi-UAV fleets, where the candidate pool would have to cover several vehicles simultaneously—a setting in which the multi-region property of the archive should be even more valuable.

9 Conclusion

This paper introduced VAST-GWO, a verified archive-guided sparse-evaluation grey wolf optimizer for optimization under limited real-evaluation budgets. The method separates cheap candidate generation—100 candidates from leader moves, success-scale alpha shells and a verified multi-region elite archive—from real evaluation, of which exactly ten uniformly drawn candidates are paid for per generation, and admits only verified information into the leaders, the archive and the rankings. On CEC2017 at $D = 50$ with 100,000 probe-audited evaluations and 30 independent runs, VAST-GWO attains the best mean rank among seven methods, significantly outperforms five of the six baselines—including a 2026 hierarchical GWO and a 2026 oppositional GWO—and leads the strongest classical improvement on the rank summary. The same ordering strengthens at $D = 100$: with an identical budget and 30 seeds, VAST-GWO attains a mean rank of 1.76 against 2.76 for the closest competitor and returns a net significant win against all six baselines under Holm correction. Controlled equal-slot ablations show that the verified multi-region archive is the performance engine: a pool of standard GWO moves alone loses 27 of the 29 functions to the full method, and replacing the archive with a single-centre sampler with identical slots, scale and acceptance rule collapses the method to 2 wins, whereas the alpha-shell route becomes optional once the archive is present. Every tested alternative scheduler—including three Q-learning policies—fails to improve on the uniform draw, and the method holds its rank under a budget five times smaller. In a 209-dimensional DEM-based 3D UAV delivery application with hard time windows, VAST-GWO is the only method feasible in all 150 runs and achieves both the best censored time to feasibility and the best paired final route quality against every baseline.

The broader lesson is that when real evaluations are the scarce resource, the structure of the candidate pool is a first-class design variable. Verifying everything that enters memory, and generating candidates from several verified regions, proved a more effective use of a fixed evaluation budget than adding scheduling complexity. We hope this encourages future work to treat the candidate pool, and the accounting of what enters the search state, as a primary object of study in population-based optimization under expensive evaluations.

Data and code availability

All optimization and application experiments, including raw per-run records, the experiment registry (every configuration and negative result), the ablation implementation, the five UAV scenarios and the analysis scripts, are available in the accompanying repository. The evaluation counts of all runs are asserted by an independent probe, and the frozen method configuration is bitwise audited against its archived implementation.

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